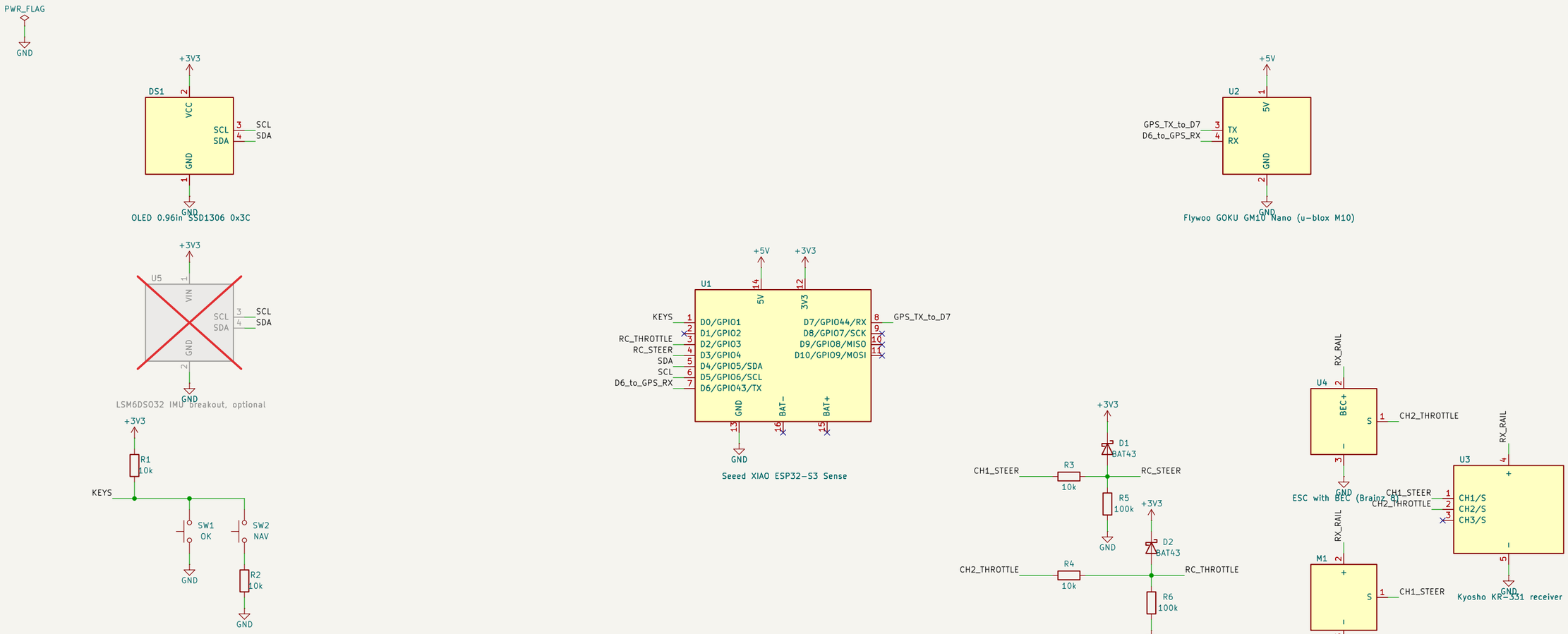


XIAO pads, USB-C up, from the top: left D0 D1 D2 D3 D4 D5 D6, right 5V GND 3V3 D10 D9 D8 D7; BAT+ and BAT- on the back. The microSD of the Sense board takes D8 CLK, D9 DAT0, D10 CMD and GPIO21 for DAT3, inside: leave D8–D10 free. D1 is spare, for the optional aux channel.

Keys: the bench's two buttons (PLAN-ESP32.md 3.1): a 10 k pull-up on D0, OK straight to ground, NAV through 10 k; 0 V, 1.62 V, and 3.12 V at rest.
The IMU is optional (DNP): on the I2C bus with the display, at 0x6A, it needs no GPIO.
The GM10 runs from the LM2940's 5 V.

Power: never the receiver rail straight into the 5V pad. C1, low ESR, and C2 soak up the servo's spikes, and the LM2940 holds 5 V for the 5V pad and the GM10. It is guaranteed from 6.25 V in; on a 5.9 V BEC its dropout, 0.2 V at most at 100 mA, leaves room, to confirm on the bench. C3 keeps it stable: 22 u or more, ESR 0.1 to 1 ohm: a low-ESR electrolytic, the EEU-FR1C101, as small tantalums run over 1 ohm. No diodes (decided 2026-10-24), so the USB is never plugged in while the car or a LiPo powers the board: the two 5 V would meet on the 5V pad.
No backup battery on BAT+/BAT- (decided 2026-10-25): the device goes off with the car, so a session is ended by hand first; one the switch cuts is closed at the next boot.

RC inputs, per channel: 10 k in series; a BAT43 to 3V3 that holds a 5 V receiver's pulse under the pin's 3.6 V; 100 k to ground that holds the pin low with its lead unplugged, so that noise is not taken for pulses. A 3.3 V receiver's high reads about 3.0 V, over 2.48 V.

The design of PLAN-ESP32.md 2.5, 2.7 and 3.1: not built yet.
To confirm on the bench: the LM2940 on a 5.9 V BEC, the RC inputs' clamps, the IMU.
Signals and keys as on bench.kicad_sch; the power and the RC inputs' protection change.

esp32-rc-telemetry

Sheet: /
File: car.kicad_sch

Title: In the car: power, protected RC inputs, two keys, IMU

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